

Fundamentals of Linear Control:

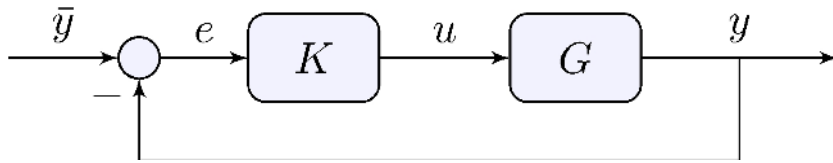
A Concise Approach

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Chapter 8: Performance and Robustness

linearcontrol.info/fundamentals

8.1 Closed-Loop Stability and Performance



Tracking performance

$$\begin{aligned} e &= S \bar{y}, \quad S = \frac{1}{1 + L}, \quad L = GK \end{aligned}$$

- Better performance needs $\|S(j\omega)\|$ small
- Because $H(s) + S(s) = 1$ $\begin{aligned} S(j\omega) \approx 0 \text{ means } H(j\omega) \approx 1 \end{aligned}$ Beware of measurement noise
- Cannot make $\|S(j\omega)\|$ small for all ω

Tracking performance

$$\begin{aligned} e &= S \bar{y}, \quad S = \frac{1}{1+L}, \quad L = GK \end{aligned}$$

High gains makes $|S|$ small

$$\begin{aligned} |L(j\omega)| &> M \gg 1 \quad \implies \quad |S(j\omega)| < \frac{1}{M-1} \end{aligned}$$

Small gains make $|S|$ one

$$\begin{aligned} |L(j\omega)| &\approx 0 \quad \implies \quad |S(j\omega)| \approx 1 \end{aligned}$$

e.g. L is strictly proper $\lim_{|s| \rightarrow \infty} |L(s)| = 0 \quad \implies \quad \lim_{|s| \rightarrow \infty} |S(s)| = 1$

Tracking performance

$$\begin{aligned} e &= S \bar{y}, & S &= \frac{1}{1+L}, & L &= GK \end{aligned}$$

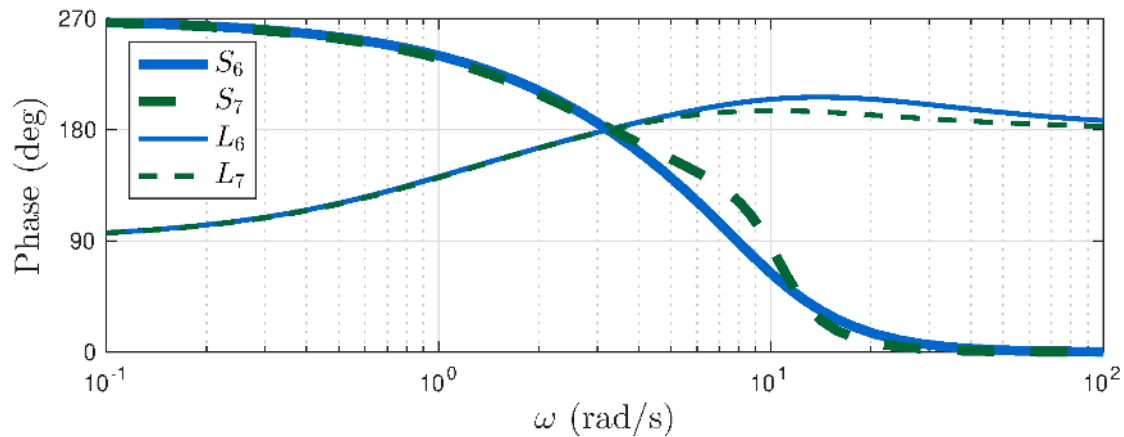
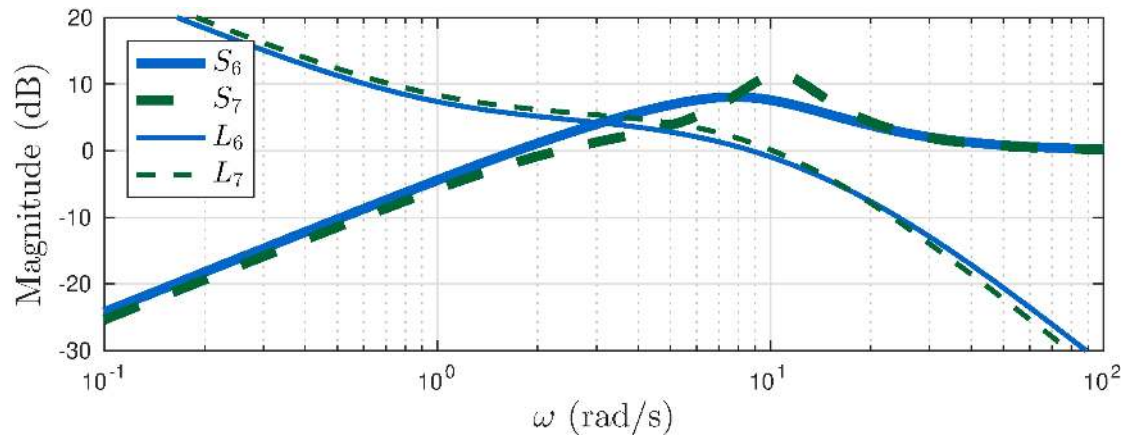
What if $|L(j\omega)| \approx 1$?

Look at the phase: $|L(j\omega)| \approx 1, \quad \angle L(j\omega) \approx \pi \implies |S(j\omega)| \approx 1$ is large

$|S|$ large makes $|S_M|$ small

$$S_M = 1/\sup_{\omega} |S(j\omega)|$$

Example: control of the simple pendulum



What happens when $\|L(j\omega)\| \approx 1$?

Bode Sensitivity Integral

Theorem 8.1:

$L(s)$ has poles p_i , $\operatorname{Re}(p_i) > 0$, $i = 1, \dots, k$. If $\lim_{|s| \rightarrow \infty} |L(s)| = \kappa$, $\kappa < \infty$ and $S = (1 + L)^{-1}$ is asymptotically stable then

$$\int_0^\infty \ln |S(j\omega)| d\omega = \int_0^\infty \ln \frac{1}{|1 + L(j\omega)|} d\omega = \pi \sum_{i=1}^k p_i - \frac{\pi \kappa}{2}$$

Main takeaways:

- Closed-loop sensitivity is on a balanced budget
- Unstable systems are hard
- Choose your battles wisely!

See book for more:

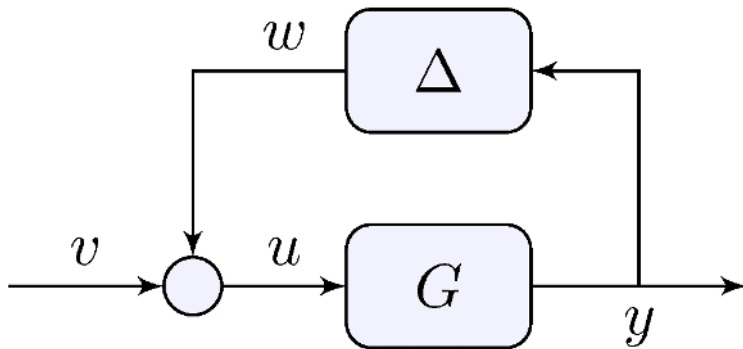
- How limited bandwidth makes things harder
- How unstable zeros amplify the effect of unstable poles

8.2 Robustness

Margins

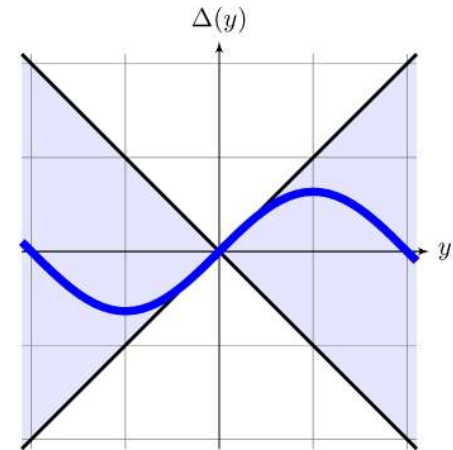
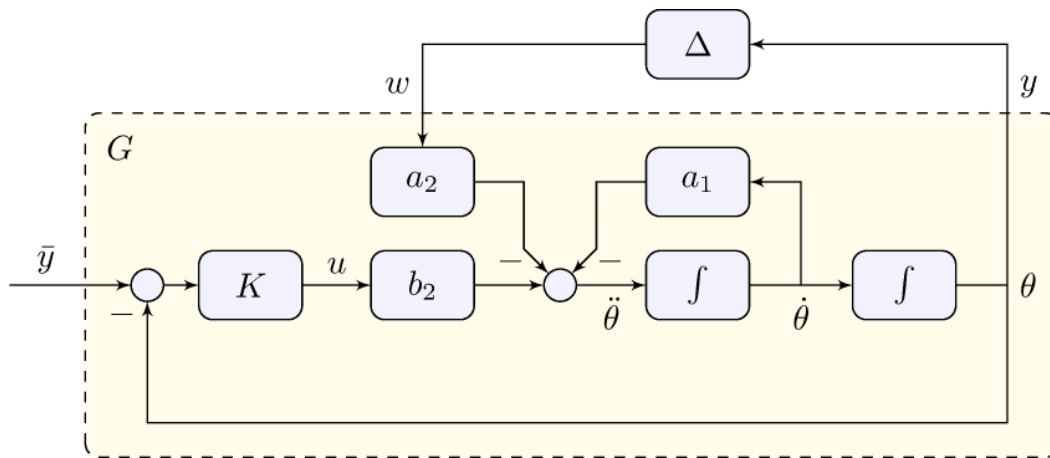
- Gain margin: robustness to changes in gain
- Phase margin: robustness to changes in phase

Robustness analysis



Robustness = asymptotic stability for all $(\Delta \in \boldsymbol{\Delta})$

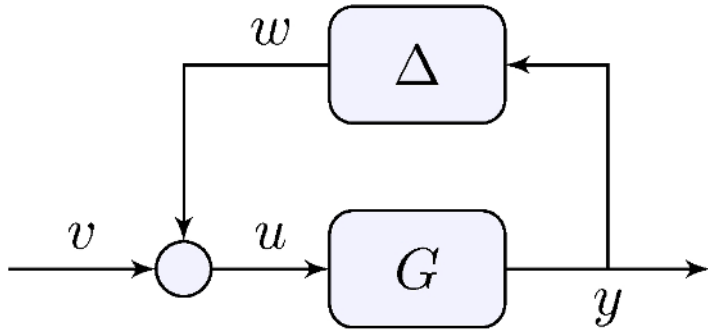
Example: control of the simple pendulum



Nonlinearity as an “uncertainty” $\Delta(y) = \sin(y)$

Uncertainty model $\Delta: \mathbb{R} \rightarrow \mathbb{R}, \quad |\Delta(y)| \leq |y|$

8.3 Small Gain



$$\begin{aligned} \boldsymbol{\Delta} &= \{ \\ \Delta: \mathbb{R} &\rightarrow \mathbb{R}, \\ \quad | \Delta(y) | &\leq |y| \} \end{aligned}$$

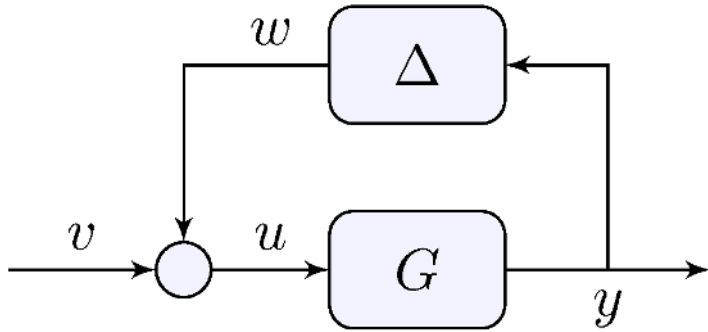
Disturbance

$$\|w\|_2 = \|\Delta(y)\|_2 \leq \|y\|_2 \quad \text{for all } \Delta \in \boldsymbol{\Delta}$$

System

$$\|y\|_2 \leq \|G\|_{\infty} (\|w\|_2 + \|v\|_2) + M \|x_0\|_2, \quad \|G\|_{\infty} = \sup_{\omega \in \mathbb{R}} |G(j\omega)|$$

How about the closed-loop?



$$\Delta: \mathbb{R} \rightarrow \mathbb{R},$$

$$|\Delta(y)| \leq |y|$$

Small gain theorem

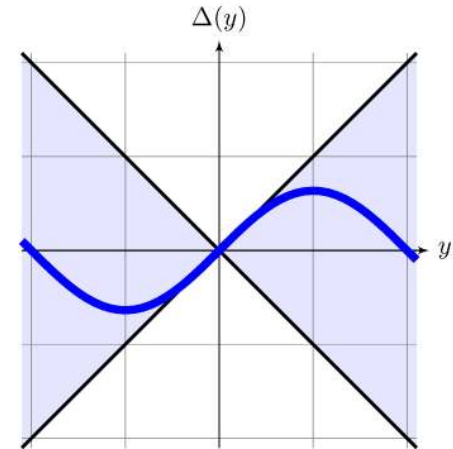
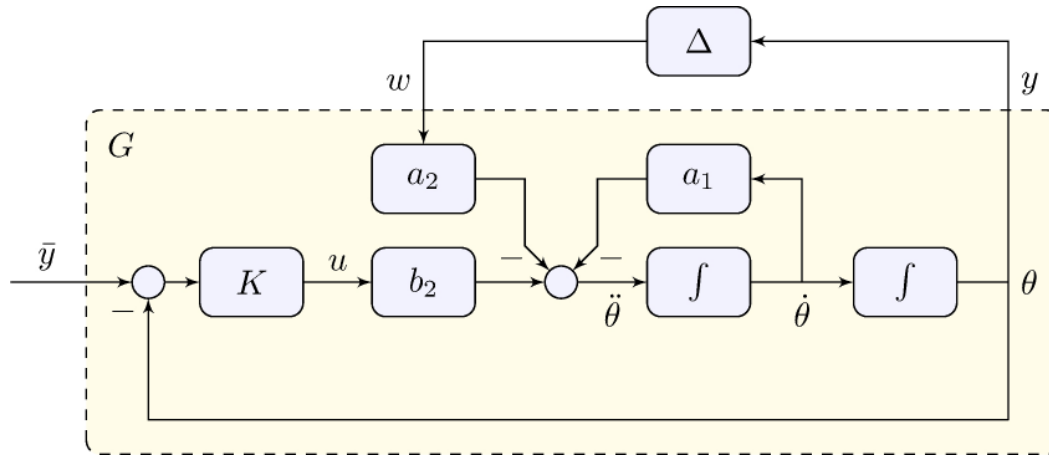
If (G) is asymptotically stable and $\|G\|_{\infty} < 1$ then

$$\|y\|_2 \leq \frac{\|G\|_{\infty}}{1 - \|G\|_{\infty}} \|v\|_2 + \frac{M}{1 - \|G\|_{\infty}} \|x_0\|_2$$

See book for more:

- Dynamic uncertainty
- Delays

8.4 Control of the Simple Pendulum - Part III



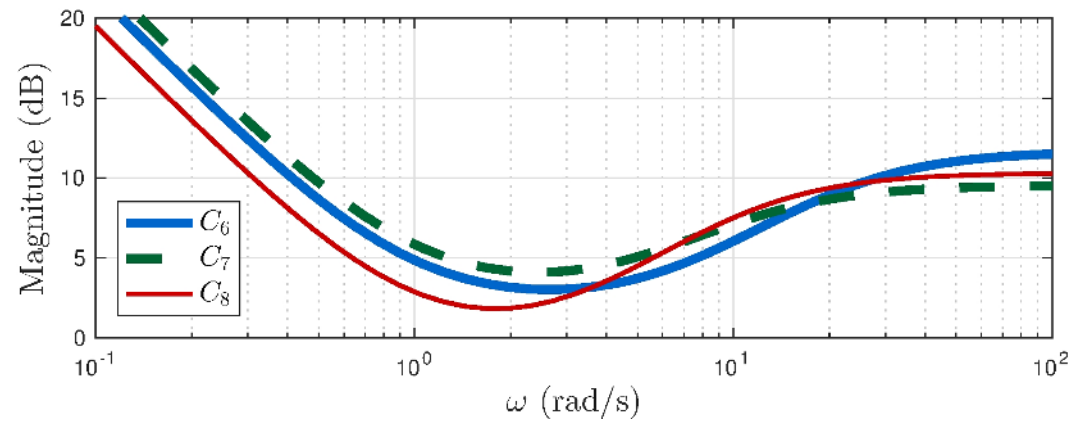
Model for robustness analysis

$$\begin{aligned} \text{\label{eq:gpendulum}} \quad G(s) &= \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, \quad F(s) = \\ &= \frac{1}{s^2 + a_1 s} \end{aligned}$$

Data

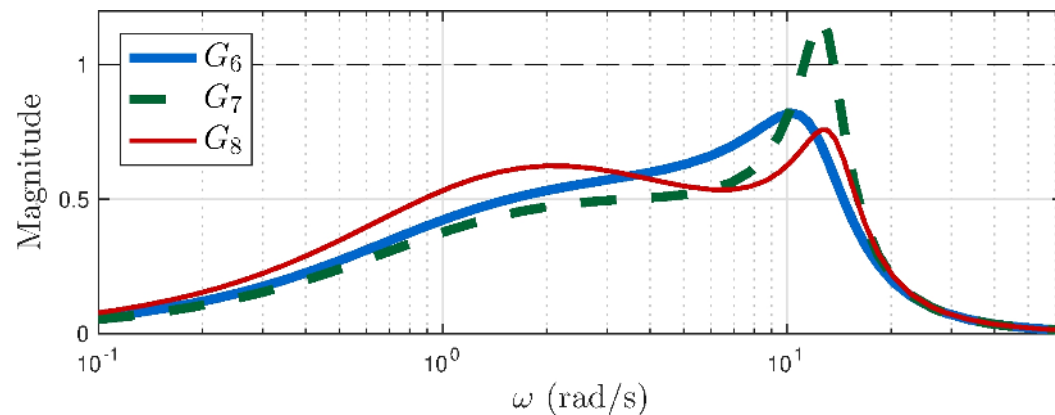
$$\begin{aligned} \text{\label{eq:ppars}} \quad a_1 &= \frac{b}{J_r} = 0, \quad a_2 = \frac{m \cdot g \cdot r}{J_r} \approx 49 \text{ s}^{-2}, \quad & b_2 &= \frac{1}{J_r} \approx 66.7 \text{ kg}^{-1} \text{ m}^{-2} \end{aligned}$$

Small gain analysis $(\bar{y} = 0)$



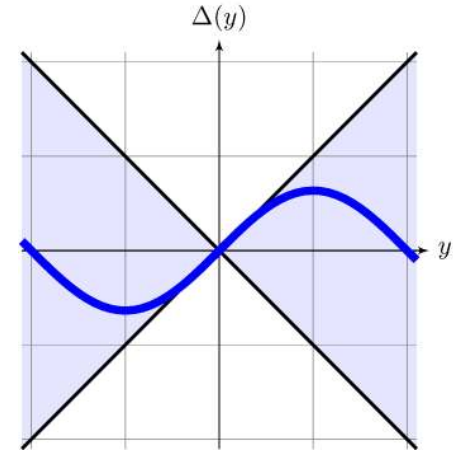
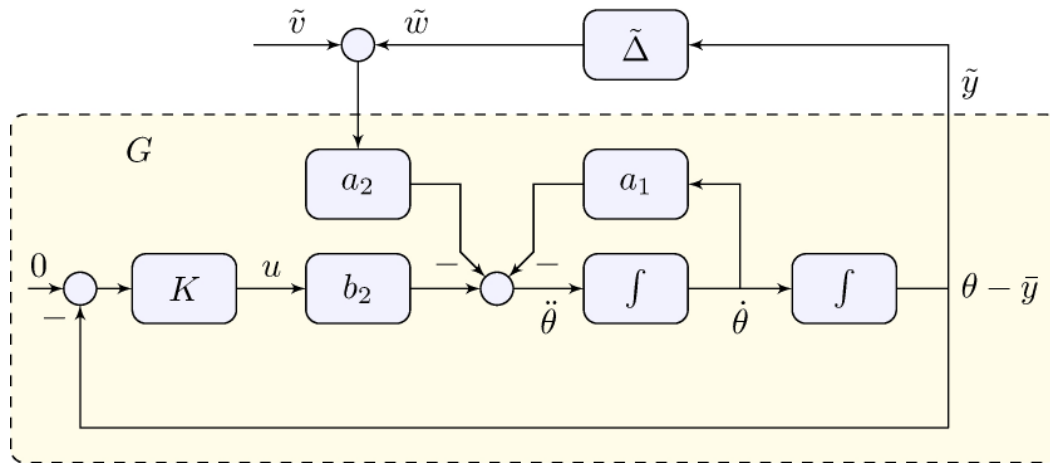
$$\begin{aligned} C_6(s) &= 3.83 \\ &\frac{(s+0.96)(s+6.86)}{s(s+21)} \\ C_7(s) &= 3 \frac{(s+1)(s+5)}{s(s+11)} \\ C_8(s) &= 3.28 \frac{(s+1)(s+3)}{s(s+10.5)} \end{aligned}$$

- C_8 : higher gains at higher frequency to lower $|G_8|_{\infty}$



$$\begin{aligned} &|G_6|_{\infty} \\ &\approx 0.82 \\ &|G_7|_{\infty} \\ &\approx 1.17 \\ &|G_8|_{\infty} \\ &\approx 0.76 \end{aligned}$$

Small gain analysis $(\|\bar{y}\| \neq 0)$



Nonlinearity

$$\begin{aligned} \tilde{\Delta}(\tilde{y}) &= \cos(\bar{y}) \sin(\tilde{y}) + \sin(\bar{y}) (\cos(\tilde{y}) - 1), & \tilde{v} &= \sin(\bar{y}) \end{aligned}$$

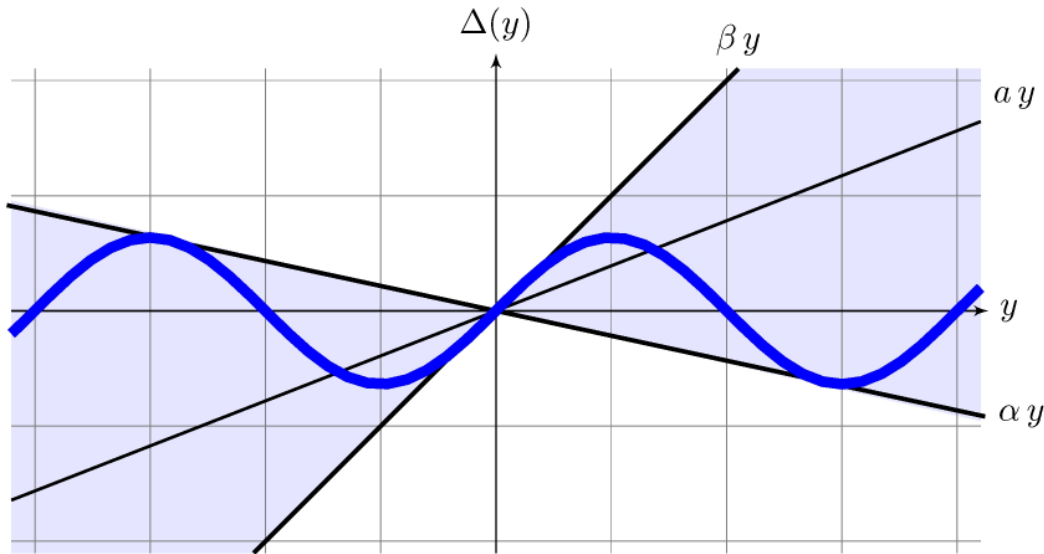
Uncertainty model

$$\tilde{\Delta} = \Delta \quad \Delta: \mathbb{R} \rightarrow \mathbb{R}, \quad \|\Delta(y)\| \leq |y|$$

Stability analysis holds also for tracking!

See book for details and more

8.5 Circle Criterion



Sector nonlinearities

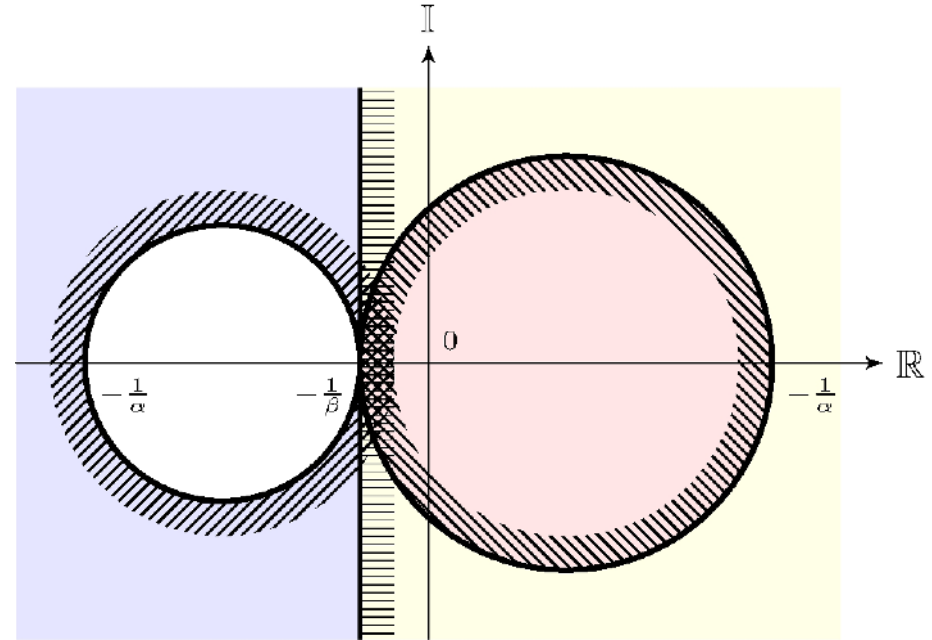
$$\Delta(y) \in [\alpha y, \beta y] \iff \Delta(y) - \alpha y \geq 0 \text{ and } \Delta(y) - \beta y \leq 0$$

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Circle Criterion

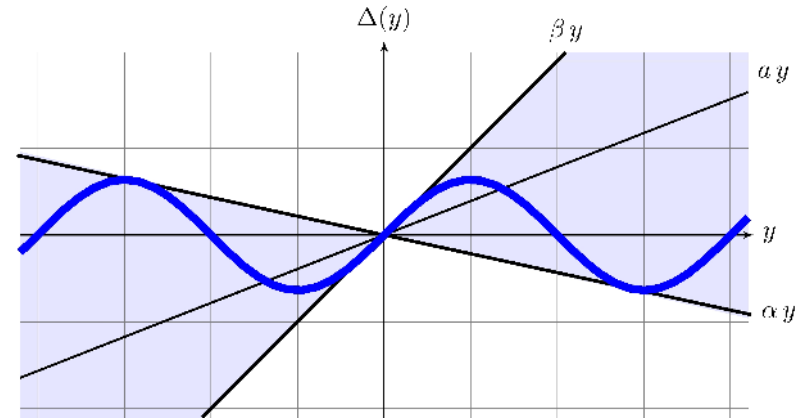
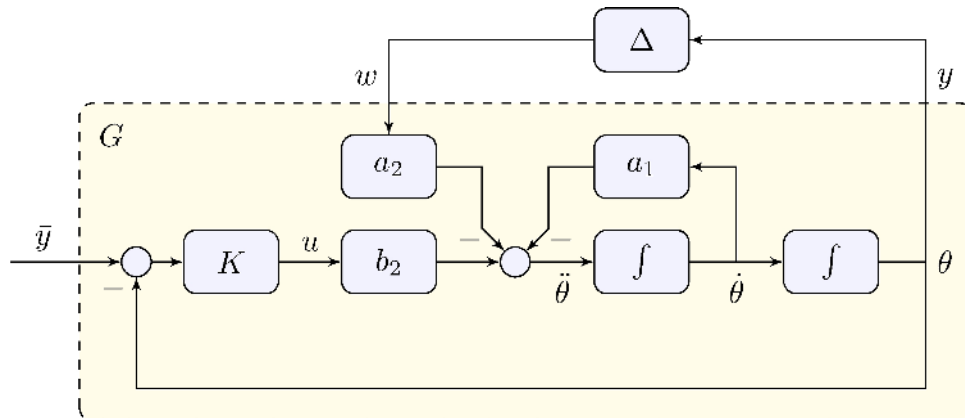
Theorem 8.2

Origin is globally asymptotically stable if either



1. $(\beta > 0 > \alpha)$, (G) is asymptotically stable and the polar plot of (G) is inside $(C(\alpha, \beta))$;
 2. $(\beta > \alpha > 0)$, the Nyquist plot of (G) encircles $(C(\alpha, \beta))$ (m) times but never enters $(C(\alpha, \beta))$.
- (m) is the number of unstable poles of (G)

Example: control of the simple pendulum



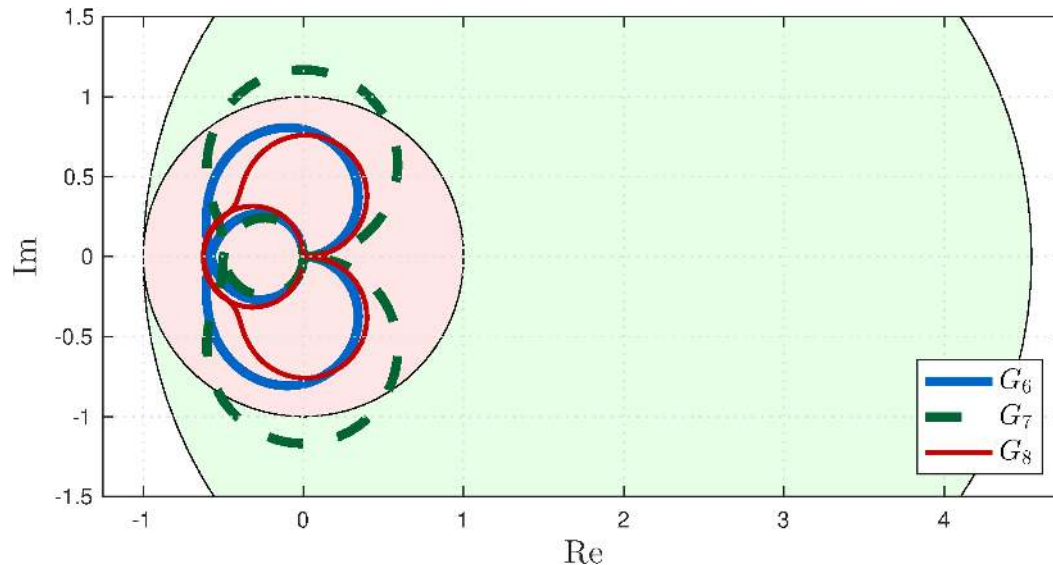
Model for robustness analysis

$$\begin{aligned} G(s) &= \frac{Y(s)}{W(s)} = \frac{-a_2 F(s)}{1 + b_2 K(s) F(s)}, & F(s) &= \frac{1}{s^2 + a_1 s}, & \alpha &= -0.22, & \beta &= 1 \end{aligned}$$

Data

$$\begin{aligned} a_1 &= \frac{b}{J_r} = 0, & a_2 &= \frac{m \cdot g \cdot r}{J_r} \approx 49 \text{ s}^{-2}, & b_2 &= \frac{1}{J_r} \approx 66.7 \text{ kg}^{-1} \text{ m}^{-2} \end{aligned}$$

Example: control of the simple pendulum



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\[\begin{align*}
C_6(s) &= 3.83 \\
&\frac{(s+0.96)(s+6.86)}{s(s+21)} \\
C_7(s) &= 3 \frac{(s+1)(s+5)}{s(s+11)} \\
C_8(s) &= 3.28 \frac{(s+1)(s+3)}{s(s+10.5)}
\end{align*}
\]

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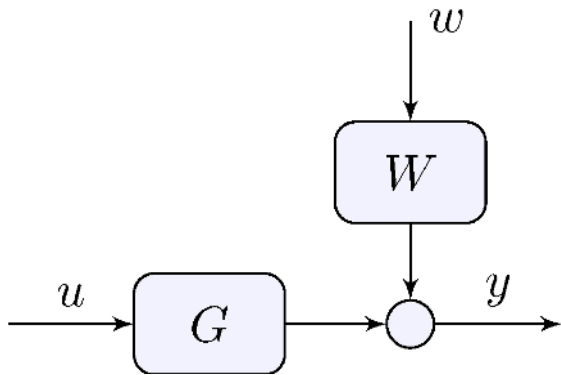
Small gain

- $(-1, 1)$
- (G_7) fails

Circle criterion

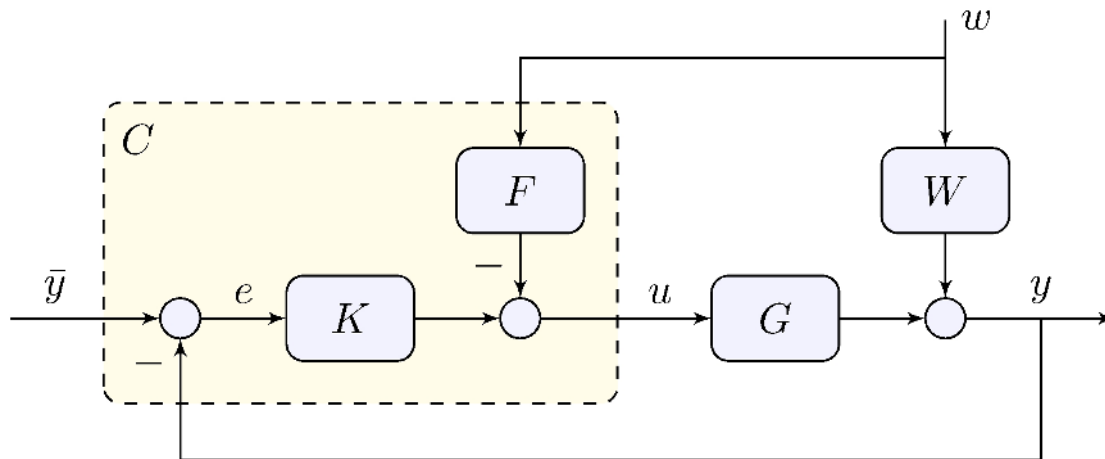
- $(-0.22, 1)$
- All controllers stabilizing

8.6 Feedforward Control and Filtering



If the “disturbance” w is known or can be measured *feedforward* can be combined with feedback

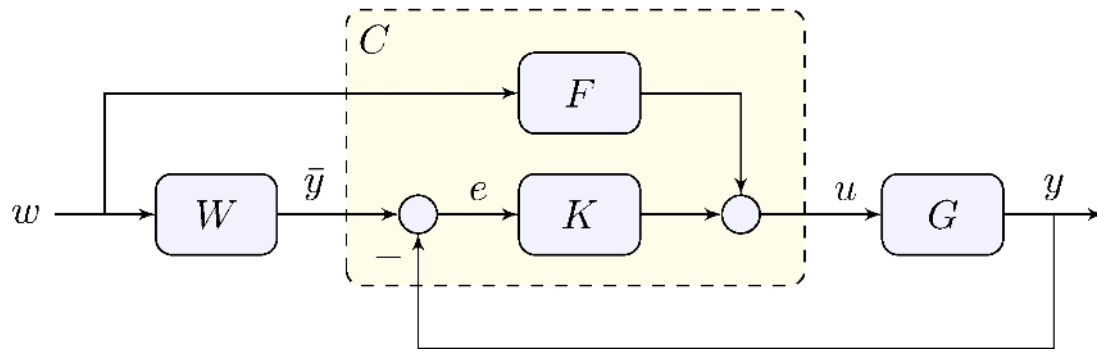
Feedforward control



Tracking error

$$\begin{aligned} e &= \bar{y} - (W - F \cdot G) \cdot S \cdot w, & S &= (1 + G \cdot K)^{-1} \end{aligned}$$

Reference feedforward control



Tracking error

$$\begin{aligned} e &= (W - F)S w, \quad S = (1 + GK)^{-1} \end{aligned}$$

Feedforward control design

Beware: since (F) is in open-loop it has to be asymptotically stable!

If possible

$$\begin{aligned} F &= G^{-1} W \quad \& \implies W = F G \end{aligned}$$

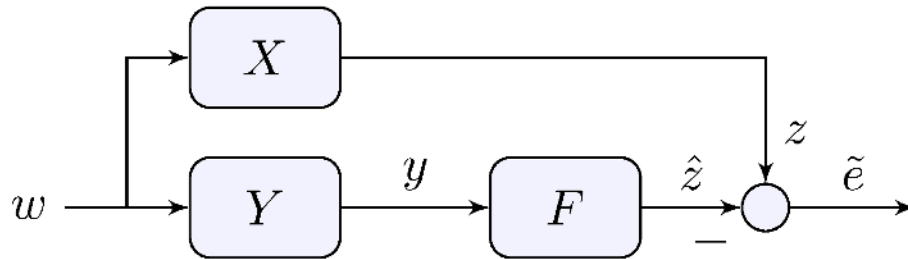
At a given frequency

$$\begin{aligned} F(j\omega) &= \lim_{s \rightarrow j\omega} G(s)^{-1} W(s) \end{aligned}$$

Optimal filter

$$\begin{aligned} \label{eq:flt} \min_{\{F\}} \| (W - F G) S \| : \quad F \text{ is asymptotically stable and realizable } \end{aligned}$$

Optimal filtering

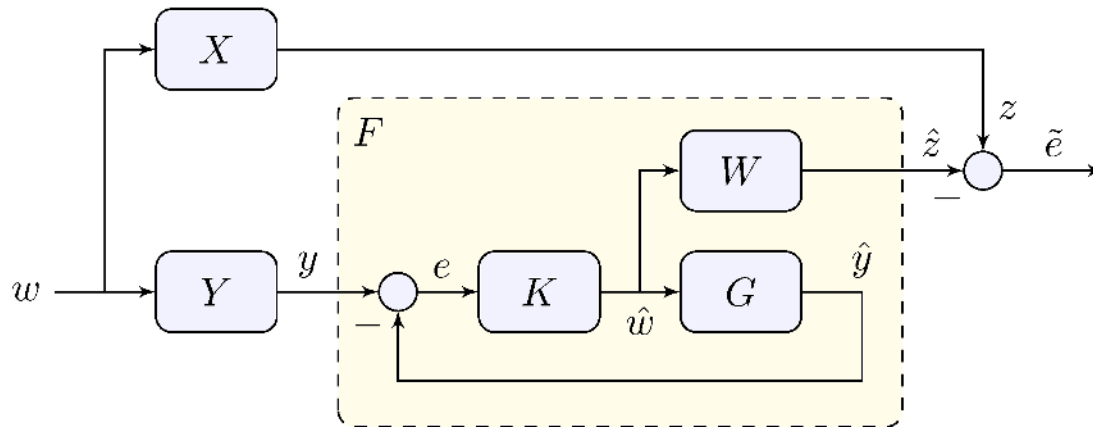


Optimization problem

$$\min_{\{X, F\}} \|X - F\| : F \text{ is asymptotically stable and realizable}$$

See book for simple examples

Optimal filtering

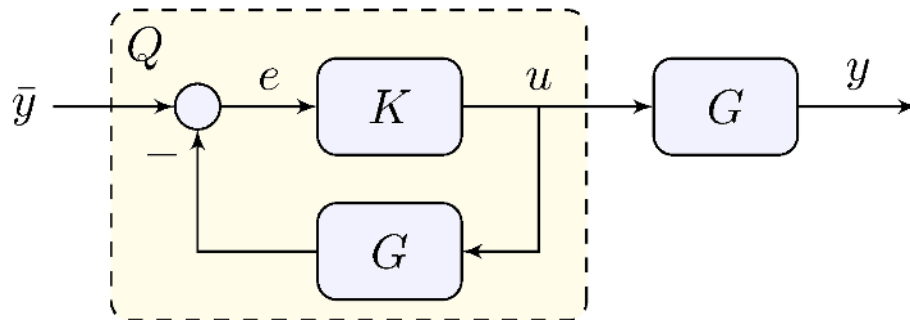


Solution via feedback

$$\begin{aligned} W &= X, & G &= Y, & \tilde{e} &= W S w, & S &= \frac{1}{1 + G \circ K} \end{aligned}$$
 e.g. (K) or (G) have a pole at zero then $\lim_{t \rightarrow \infty} \tilde{e}(t) = 0$

This is similar to a *Kalman filter*

All Stabilizing Controllers



Assumption: (G) is asymptotically stable

$$K \text{ is stabilizing} \implies Q = \frac{K}{1 + K G} \text{ is asymptotically stable} \\
Q \text{ is asymptotically stable} \implies K = \frac{Q}{1 - Q G} \text{ is stabilizing}$$

A non-trivial version exists in which (G) does not need to be asymptotically stable

Learn more

- [Book website](#)

